Two EndRing recipient mechanisms

Proposal and adversarial evaluation

GOAL-SSI-2fa82a · BATCH-d92210 · TASK-20261013-9e1246 · 2026-10-13

Result

This round did **not** build a secure PKE. It proposed and independently evaluated two materially different interfaces. M1, LABELLED-NORM-HOM-LIFT, remains a construction gap: its Hom-lattice lift, complete norm-fiber enumeration, fiber-correctness bound, and attacker-to-complete-EndRing extractor are absent. M2, simultaneous full-module conjugation, is rejected for the submitted syntax: public linear algebra recovers the normalized conjugator used by its mask.

Novelty, QPT/QROM security, a correct-direction reduction to complete EndRing, and NIST Category 5 remain **unestablished**. No parameter set was selected. Scientific, implementation, formalizer, benchmark, and parameter-selection runs were all zero.

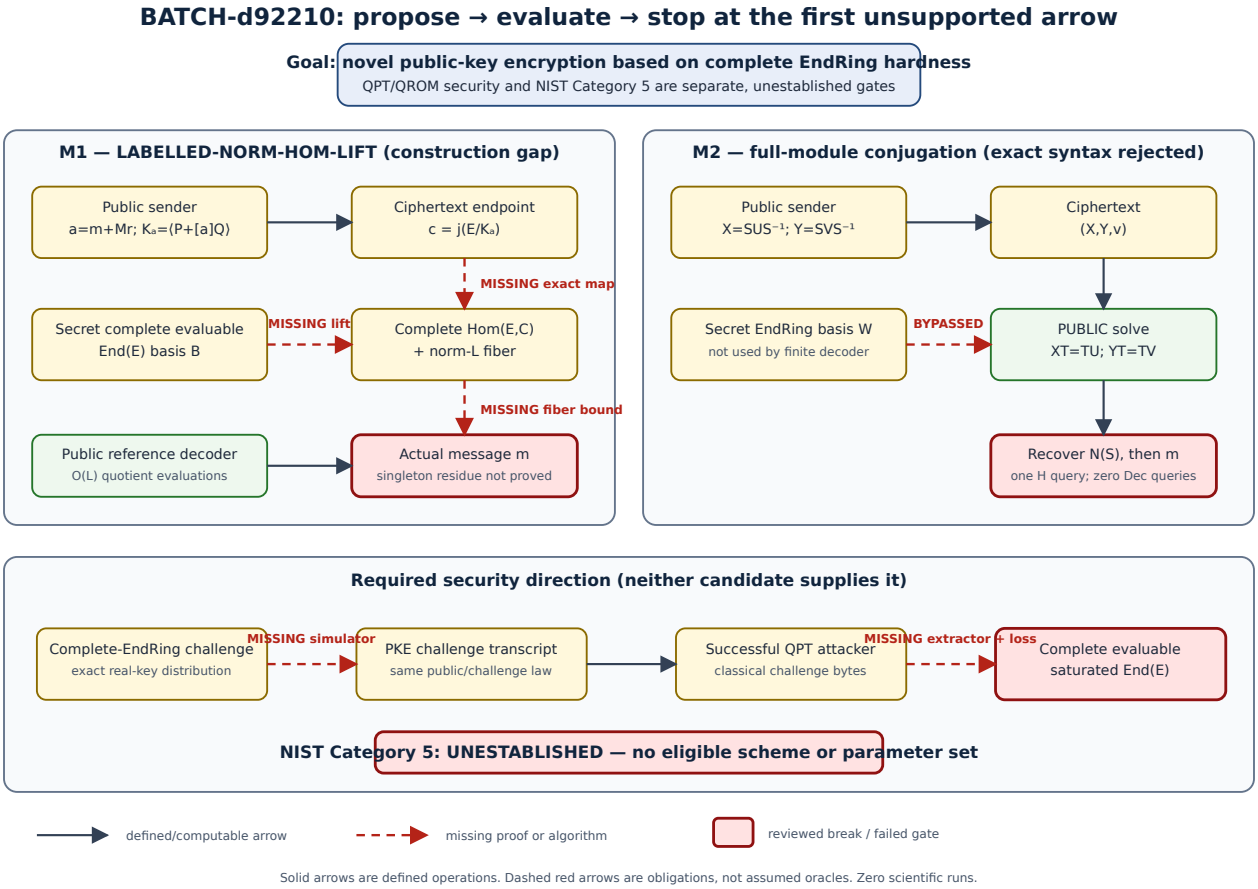


Figure 1: Two attempted recipient mechanisms and the exact missing or broken arrows. Solid arrows are defined operations; dashed red arrows are obligations, not assumed oracles.

Frozen propose/evaluate method

The frozen method admitted at most two materially distinct recipient operations, required O1–O6 packets, and assigned independent reviews: J1/J2 correctness and novelty, J3/J4 simulation and extraction, and J5/J6 quantum and Category-5 claims. The composition copied owner verdicts without voting:

Joint	Verdict	Narrow scope
J1	breaks	M1 inconclusive; M2 does not require a recipient secret.
J2	inconclusive	Operation changes exist, but novelty is unverified.
J3	inconclusive	Neither exact-law simulator is complete.

J4	breaks	Neither attacker-to-complete-EndRing reduction is typed.
J5	breaks	M1 lacks a QPT theorem; M2 has public passive recovery.
J6	inconclusive	Complete costs and a Category-5 case are unavailable.

M1 — LABELLED-NORM-HOM-LIFT

Proposed syntax

For a public supersingular curve E over the quadratic field, public basis (P,Q) of the L-torsion, and M dividing L, encryption samples r, sets a=m+Mr and the cyclic kernel generated by P+[a]Q, and emits the j-invariant of the quotient. The public sender needs no ring. The intended private operation would use a complete evaluable rank-four basis of End(E) to construct all of Hom(E,C), enumerate all norm-L maps, and recover one residue modulo M.

Reviewed gaps

Correctness is exact only conditionally: if F(c) is the set of labels reaching c, the reference decoder returns m exactly when every a in F(c) has residue m modulo M. No negligible cross-message fiber bound is supplied.

Knowing End(E) does not itself give a connecting map, a complete Hom(E,C) basis, principalization, or bounded norm enumeration. The exact reference decoder publicly enumerates all L labels, so polynomial L supplies no recipient asymmetry. A single norm-L Hom witness reveals at most one label and is not a complete-ring separator.

The required reduction must map a successful QPT labelled-quotient distinguisher to a complete evaluable saturated basis of End(E). No exact-law challenge simulator, extractor, loss, or adaptive-OneEnd bridge is supplied. O1–O6 remain open. M1 is a construction gap, not a PKE or impossibility theorem.

M2 — simultaneous full-module conjugation

Proposed syntax

Public U,V generate the full two-by-two matrix algebra over Fℓ. The sender samples an invertible matrix S, publishes its simultaneous conjugates X and Y, and masks m with a hash of the normalized projective class N(S).

Public recovery

Solve the public equations XT=TU and YT=TV. For an honest transcript write T=SC. Then C commutes with U and V. Their generated algebra is the full matrix algebra, whose centralizer is scalar, so the solution space is exactly the scalar multiples of S. Every nonzero solution has the same normalization N(S).

A classical passive adversary, hence also a valid QPT adversary, recovers the exact hash input. One ordinary hash query and zero decryption queries recover the selected message. For distinct challenge messages the advantage is 1/2. Conditional actual-message correctness is discharged, but it is public correctness; O4 is contradicted for this syntax. An integral lift cannot hide a pad already fixed by public matrices.

Obligation ledger

Obligation	M1	M2
O1 exact hard distribution	open	open
O2 total polynomial algorithms	open	open
O3 actual-message correctness	open	conditional discharge
O4 complete-ring necessity	open	contradicted
O5 attacker → complete EndRing	open	open; gate breaks
O6 costs / security evidence	open	open; Cat-5 rejected

There are zero candidates eligible for implementation, benchmark, parameter selection, or a security claim.

Quantum and Category-5 boundary

NIST Category 5 compares applicable attack resources with key search on a 256-bit block cipher. The retained criteria page gives preliminary references of 2-to-the-272 classical gates and 2-to-the-298 divided by MAXDEPTH quantum gates. These are accounting references, not certification of a field size.

A future candidate needs a complete polynomial scheme and exact real-key law; a QPT/QROM game with query, width, depth, gate, memory, data, target, preprocessing, and success accounting; an explicit advantage-to-EndRing-success inequality with loss and additive error; and independently reproduced classical and quantum attack margins. M1 has no eligible algorithms, losses, or parameters. M2 has a polynomial public break. Bit length alone is not Category-5 evidence.

Novelty, procedure, and scope limits

Retained sources include NIST primary materials and abstract- or metadata-level records for Séta (ePrint 2019/1291), SiGamal (2020/613), LIT-SiGamal (2024/521), and supersingular key-updatable PKE (2020/1593). Full normalized constructions were unavailable for every neighbor. These sources identify comparison targets, not novelty; neither candidate establishes nonduplication.

The independence checker exited 1 with eight diagnostics. Two review files omit the nested task identifier in their attestations, so it does not recognize J1/J2, J5/J6, or blind-rederivation ownership; J7 belongs to later document review. That administrative failure is preserved and is not called a pass.

No experiment was run. Symbolic derivations are not timing measurements or attack lower bounds. The M2 result applies to the exact submitted syntax. M1′s missing algorithm is not a proof of nonexistence.

Canonical graph check: knowledge/frontiers/ecdlp contains no affected graph. Historical research reports are immutable snapshots and were not edited.

Exact next boundary

No automatic successor is authorized. A later proposal must jointly supply public encryption without the ring, an exact efficient operation available from the complete ring but not public finite data or a smaller witness, actual-message correctness on the exact distribution, and a QPT-compatible attacker-to-complete-EndRing reduction with explicit losses. A hash wrapper, authentication tag, or more evaluation points does not by itself change the failed recipient relation.

Evidence map

- R1: TASK-20261013-44f353 composition, frozen by TASK-20261013-b6cc59.
- R2: TASK-20261013-962e06, J1/J2 review.
- R3: TASK-20261013-235da3, J3/J4 review.
- R4: TASK-20261013-0e9c61, blind J5/J6 review.
- P1/P2: TASK-20261013-f8430e and TASK-20261013-6bd28a candidate packets.
- S1: NIST Post-Quantum Cryptography Security Evaluation Criteria and 2016 call, retained local captures.